

## 1. General Description

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The EMS4100 is a high-speed bidirectional passive switch in mux or demux configurations suited for USB Type-C™ application supporting USB 3.1 Gen 1 and Gen 2 data rates. Based on control pin SEL, the device provides switching on differential channels between Port B or Port C to Port A. The EMS4100 is a generic analog differential passive switch that can work for any high-speed interface applications requiring a common mode voltage range of 0 to 2 V and differential signaling with differential amplitude up to 1800 mVpp. It employs adaptive tracking that ensures the channel remains unchanged for the entire common mode voltage range. Excellent dynamic characteristics of the device allow high-speed switching with minimum attenuation to the signal eye diagram with very little added jitter. It consumes <2mW of power when operational and has a shutdown mode exercisable by EN pin resulting <20uW.

## 2. Features and Benefits

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- 2-Differential Channel 1:2/2:1 Mux/DeMux
- USB 3.1 Super Speed 10Gbps Switch
- High Bandwidth: 3.5GHz @ -3dB BW
- Supports both AC coupled and DC coupled signals
- Isolation: -24dB @ 1.25 Gbps
- Crosstalk: -34dB @ 1.25 Gbps
- ESD Tolerance: 2kV HBM
- Low bit-to-bit skew, Bidirectional
- Wide VCC Operating Range: 1.5v ~ 5.0v
- Small Packaging, QFN1.8x2.6-16L and QFN2x3-18L

## 3.Applications

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- USB Type-C Ecosystem
- Desktop and Notebook PCs
- Server/Storage Area Networks
- PCI Express Backplanes
- Shared I/O Ports
- FPD LinkII and FPD LinkIII Switching

## EMS4100

Two-Channel Differential 2:1/1:2 USB 3.1 Super Speed 10Gbps Mux/DeMux

## 4. Ordering Information

Table 1. Ordering information

| Type number | Topside marking | Package        |                                                              |          |
|-------------|-----------------|----------------|--------------------------------------------------------------|----------|
|             |                 | Name           | Description                                                  | Quantity |
| EMS4100Q    | ASW10           | QFN1.8×2.6-16L | QFN package, 16 pins<br>1.8 mm × 2.6 mm; 0.6 mm (Max) height | 3000     |
| EMS4100N    | A340AXYW        | QFN2x3-18L     | QFN package, 18 pins<br>2 mm × 3 mm; 0.4 mm (Max) height     | 3000     |

### MARKING INFORMATION

NOTE: A340AXYW =Product code and Data Code for EMS4100N

A340AX Y W

└── Data Code -Week

└── Data Code -Year

└── Product Code

## 5. Block Diagram

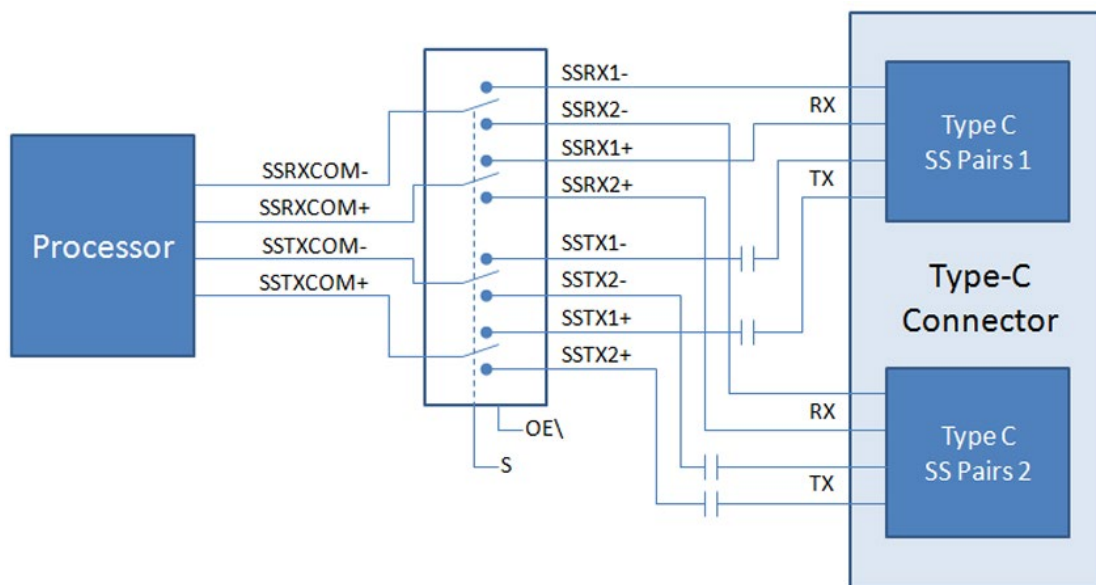


Fig 1. Block diagram

## EMS4100

Two-Channel Differential 2:1/1:2 USB 3.1 Super Speed 10Gbps Mux/DeMux

# 6. Pinning Information

## 6.1. Pin map

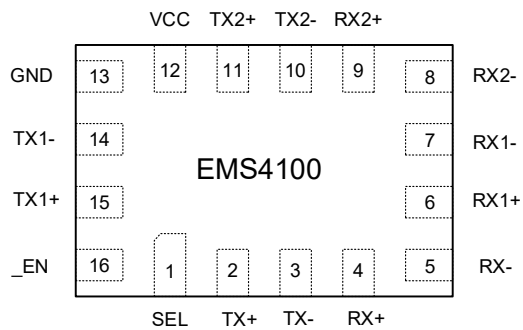


Fig 2. Top view pin configuration QFN1.8x2.6-16L

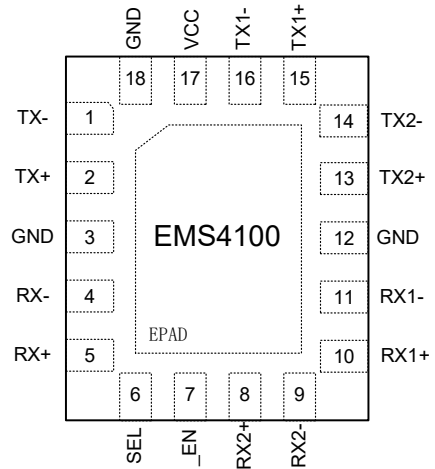


Fig 3. Top view pin configuration QFN2x3-18L

## EMS4100

### Two-Channel Differential 2:1/1:2 USB 3.1 Super Speed 10Gbps Mux/DeMux

## 6.2. Pin description

Table 2. Pin description

| Symbol | Pin            |            | Signal Type | Description             |
|--------|----------------|------------|-------------|-------------------------|
|        | QFN1.8x2.6-16L | QFN2x3-18L |             |                         |
| SEL    | 1              | 6          | I           | Switch logic control    |
| TX+    | 2              | 2          | I/O         | Super Speed TX+ Common  |
| TX-    | 3              | 1          | I/O         | Super Speed TX- Common  |
| RX+    | 4              | 5          | I/O         | Super Speed RX+ Common  |
| RX-    | 5              | 4          | I/O         | Super Speed RX- Common  |
| RX1+   | 6              | 10         | I/O         | Super Speed RX1+        |
| RX1-   | 7              | 11         | I/O         | Super Speed RX1-        |
| RX2-   | 8              | 9          | I/O         | Super Speed RX2-        |
| RX2+   | 9              | 8          | I/O         | Super Speed RX2+        |
| TX2-   | 10             | 14         | I/O         | Super Speed TX2-        |
| TX2+   | 11             | 13         | I/O         | Super Speed TX2+        |
| VCC    | 12             | 17         | Power       | Supply Voltage          |
| GND    | 13             | 3, 12, 18  | GND         | Ground                  |
| TX1-   | 14             | 16         | I/O         | Super Speed TX1-        |
| TX1+   | 15             | 15         | I/O         | Super Speed TX1+        |
| _EN    | 16             | 7          | I           | Chip Enable, Active Low |
| EPAD   |                |            |             | Ground or Floating      |

## 7.Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = Don't care; Z = high-impedance.

| Input |     | Switch               |
|-------|-----|----------------------|
| _EN   | SEL |                      |
| H     | X   | Switch off           |
| L     | L   | RX to RX1, TX to TX1 |
| L     | H   | RX to RX2, TX to TX2 |

## 8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol             | Parameter                                      | Conditions                            | Min  | Max             | Unit |
|--------------------|------------------------------------------------|---------------------------------------|------|-----------------|------|
| V <sub>CC</sub>    | supply voltage                                 |                                       | -0.5 | 5.5             | V    |
| V <sub>I</sub>     | DC input voltage                               | [1]                                   | -0.5 | V <sub>CC</sub> | V    |
| V <sub>TX/RX</sub> | Super Speed Data Channel TX / RX input voltage | [1]                                   | -0.5 | 3.8             | V    |
| I <sub>O</sub>     | DC output current                              |                                       |      | 50              | mA   |
| P <sub>tot</sub>   | total power dissipation                        | T <sub>amb</sub> = -40 °C to + 125 °C |      | 300             | mW   |
| T <sub>J</sub>     | Junction temperature                           |                                       | -40  | 125             | °C   |
| T <sub>stg</sub>   | storage temperature                            |                                       | -65  | 150             | °C   |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## EMS4100

### Two-Channel Differential 2:1/1:2 USB 3.1 Super Speed 10Gbps Mux/DeMux

## 9. Electrical Characteristics

**Table 5. Static characteristics**

$T_a=25^{\circ}\text{C}$ ,  $V_{CC}=1.8\text{V}$ , unless otherwise specified

| Symbol                    | Parameter                          | Conditions                                                 | Min  | Typ  | Max  | Unit          |
|---------------------------|------------------------------------|------------------------------------------------------------|------|------|------|---------------|
| <b>POWER SUPPLY</b>       |                                    |                                                            |      |      |      |               |
| $I_Q$                     | VCC Quiescent Current              | SEL=0 or VCC, _EN=0                                        |      | 28   |      | $\mu\text{A}$ |
| $I_{PD}$                  | Power-down Current                 | SEL=0 or VCC, _EN=VCC                                      |      |      | 1    | $\mu\text{A}$ |
| <b>DC CHARACTERISTICS</b> |                                    |                                                            |      |      |      |               |
| $V_{IH}$                  | Input logic high                   | VCC=1.8~4.5V                                               | 1.6  |      |      | V             |
| $V_{IL}$                  | Input logic low                    | VCC=1.8~4.5V                                               |      |      | 0.4  | V             |
| $R_{UP}$                  | _EN Internal pull-up resistor      |                                                            |      | 2    |      | M $\Omega$    |
| $R_{DN}$                  | SEL Internal pull-down resistor    |                                                            |      | 2    |      | M $\Omega$    |
| $R_{ON\_HS}$              | On-Resistance for TX/RX            | $V_{IS}=0.2\text{V}$ $I_{ON}=8\text{mA}$                   |      | 6.7  | 8    | $\Omega$      |
| $R_{FLAT\_LP}$            | $R_{ON}$ Flatness for TX/RX        | $V_{IS}=0$ to 1.2V $I_{ON}=8\text{mA}$                     |      | 0.8  | 1    | $\Omega$      |
| $R_{FLAT\_LP}$            | $R_{ON}$ Flatness for TX/RX        | $V_{IS}=0$ to 0.2V $I_{ON}=8\text{mA}$                     |      | 0.2  | 0.3  | $\Omega$      |
| $R_{MATCH}$               | $R_{ON}$ Matching Between Channels | $V_{IS}=0$ to 1.2V $I_{ON}=8\text{mA}$                     |      | 0.1  |      | $\Omega$      |
| $I_{OFF}$                 | Switch Off Leakage Current         | _EN=VCC, Tx, Rx =VCC<br>TX1, TX2, RX1, RX2=0               | -0.5 |      | 0.5  | $\mu\text{A}$ |
| <b>AC CHARACTERISTICS</b> |                                    |                                                            |      |      |      |               |
| $t_{EN}$                  | Enable Time _EN to Output          | $R_L=50\Omega$ $C_L=0\text{pF}$ $V_{IS}=0.6\text{V}$       |      | 80   | 150  | $\mu\text{S}$ |
| $t_{DIS}$                 | Disable Time _EN to Output         | $R_L=50\Omega$ $C_L=0\text{pF}$ $V_{IS}=0.6\text{V}$       |      | 40   | 250  | nS            |
| $t_{ON}$                  | Turn-On Time SEL to Output         | $R_L=50\Omega$ $C_L=0\text{pF}$ $V_{IS}=0.6\text{V}$       |      | 400  | 1200 | nS            |
| $t_{OFF}$                 | Turn-Off Time SEL to Output        | $R_L=50\Omega$ $C_L=0\text{pF}$ $V_{IS}=0.6\text{V}$       |      | 130  | 800  | nS            |
| $t_{BBM}$                 | Break-Before-Make Time             | $R_L=50\Omega$ $C_L=0\text{pF}$ $V_{IS}=0.6\text{V}$       |      | 250  | 500  | nS            |
| $t_{PD}$                  | Propagation Delay                  | $R_L=50\Omega$ $C_L=0\text{pF}$ $V_{IS}=0.6\text{V}$       |      | 0.25 |      | nS            |
| Off                       | Off Isolation                      | $R_L=50\Omega$ $f=1.2\text{GHz}$ $V_{IS}=0.2\text{V}_{PP}$ |      | -27  |      | dB            |

## EMS4100

### Two-Channel Differential 2:1/1:2 USB 3.1 Super Speed 10Gbps Mux/DeMux

|                    |                        |                                                           |  |     |  |     |
|--------------------|------------------------|-----------------------------------------------------------|--|-----|--|-----|
| $X_{TALK}$         | Crosstalk              | $R_L = 50\Omega$ $f = 1.2\text{GHz}$ $V_{IS} = 0.2V_{pp}$ |  | -43 |  | dB  |
| $BW_{-3dB}$        | -3dB Bandwidth         | $R_L = 50\Omega$ $C_L = 0pF$ Signal 0dBm                  |  | 3.5 |  | GHz |
| <b>CAPACITANCE</b> |                        |                                                           |  |     |  |     |
| $C_{ON}$           | Switch On Capacitance  | $V_{Bias} = 0.2V$ , $f = 1.5\text{GHz}$                   |  | 1.5 |  | pF  |
| $C_{OFF}$          | Switch Off Capacitance | $V_{Bias} = 0.2V$ , $f = 1.5\text{GHz}$                   |  | 1.0 |  | pF  |

- [1] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.  
 [2] RON matching between channels is calculated by subtracting the channel with the lowest max Ron value from the channel with the highest max Ron value.  
 [3] Crosstalk is inversely proportional to source impedance.

## 10. Application Information

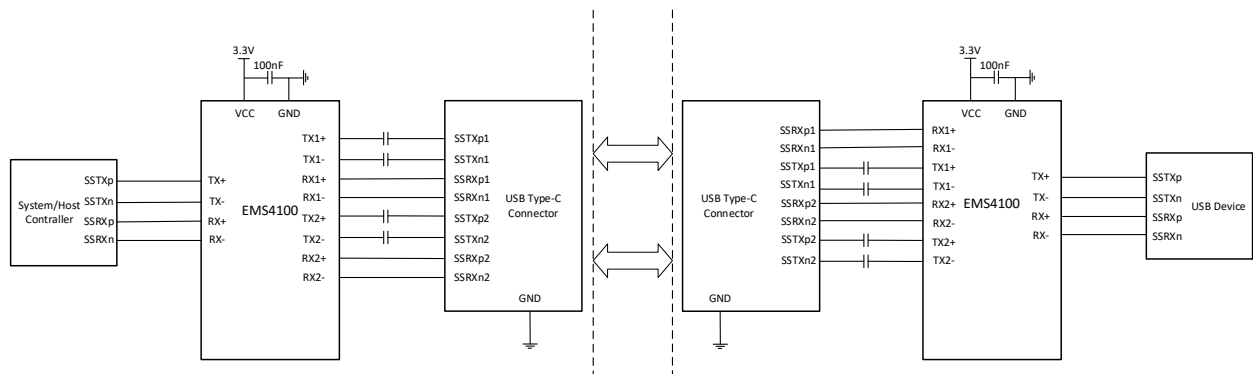


Fig 4. AC Coupling Capacitors for USB Type-C

## 11. PCB Layout

### 11.1. Layout Example

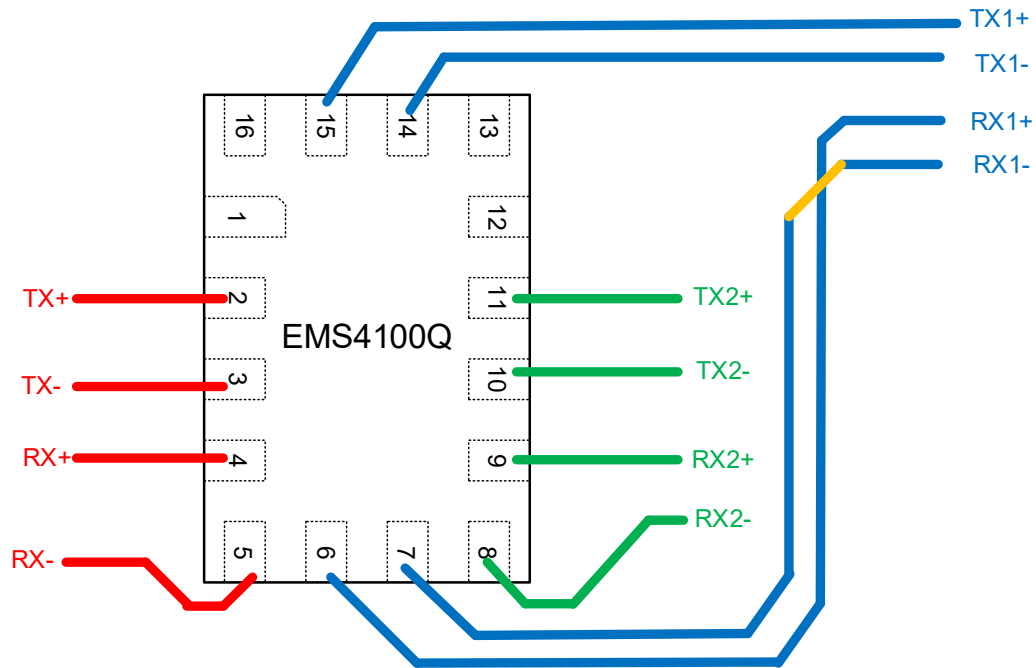


Fig 5. Layout example 1

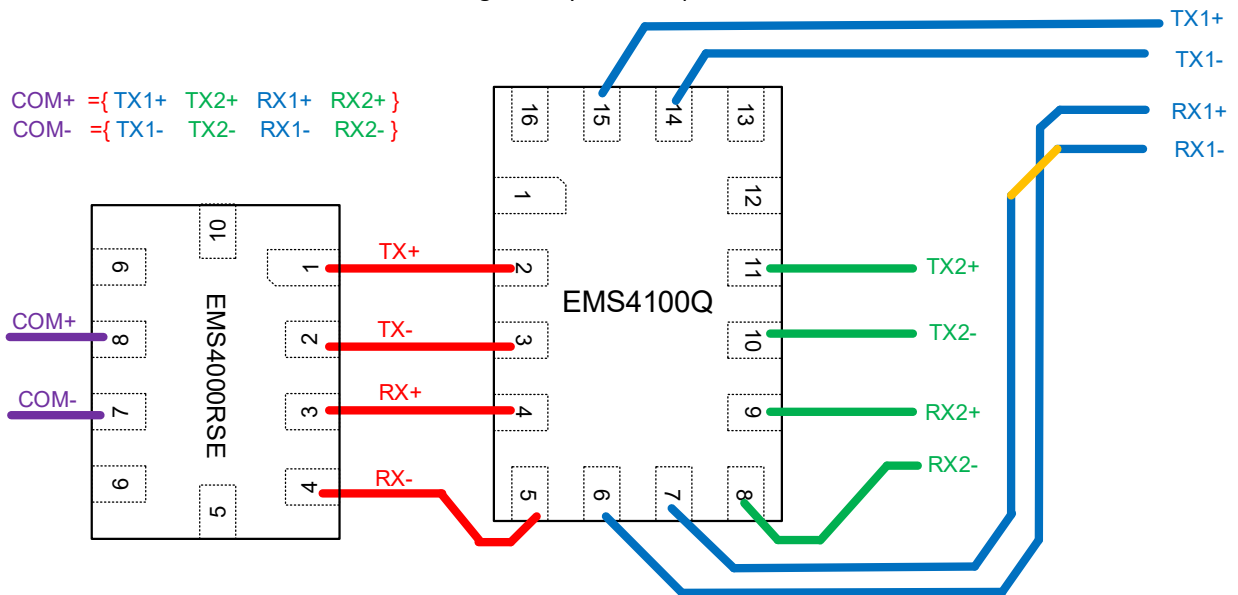
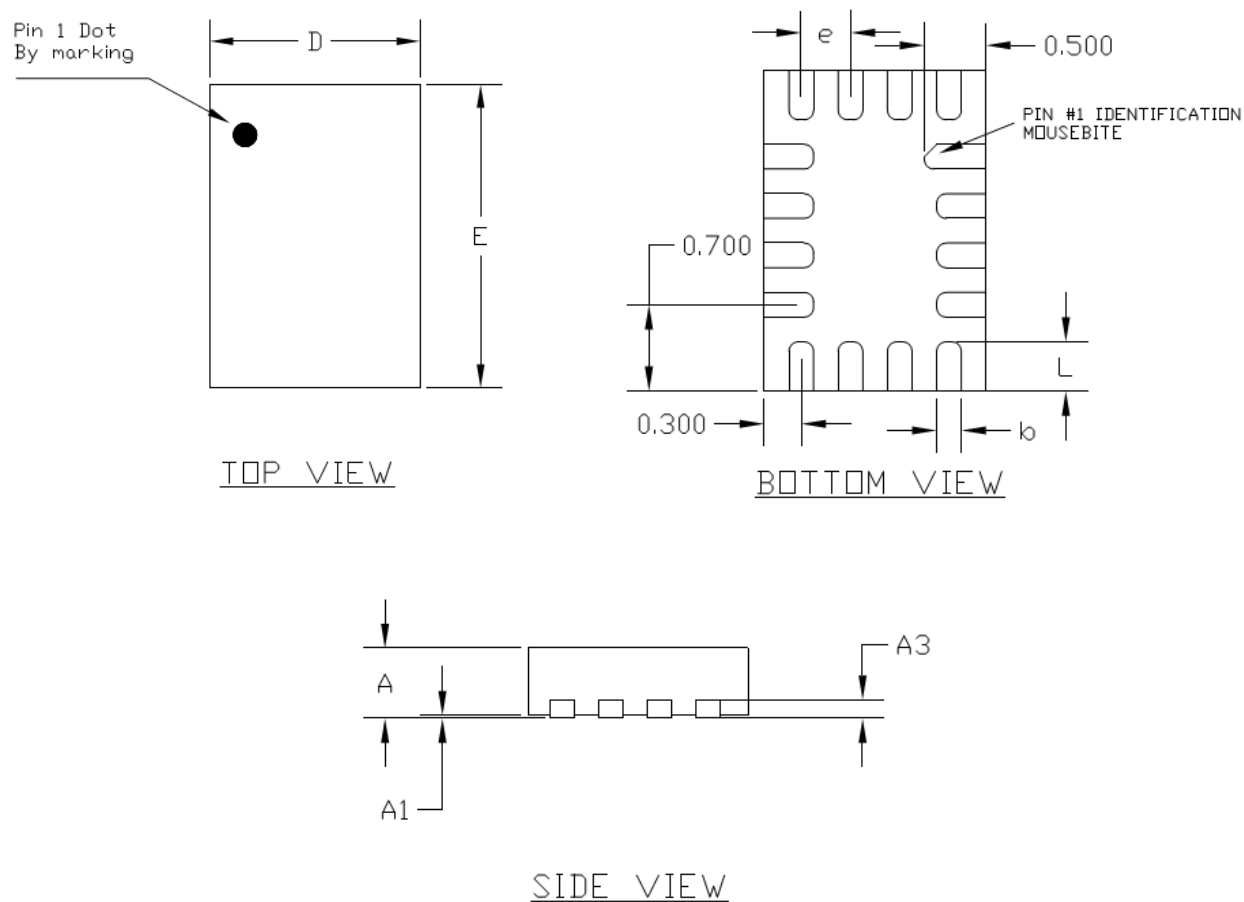


Fig 6. Layout example 2

## 12. Package Outline

### QFN1.8x2.6-16L

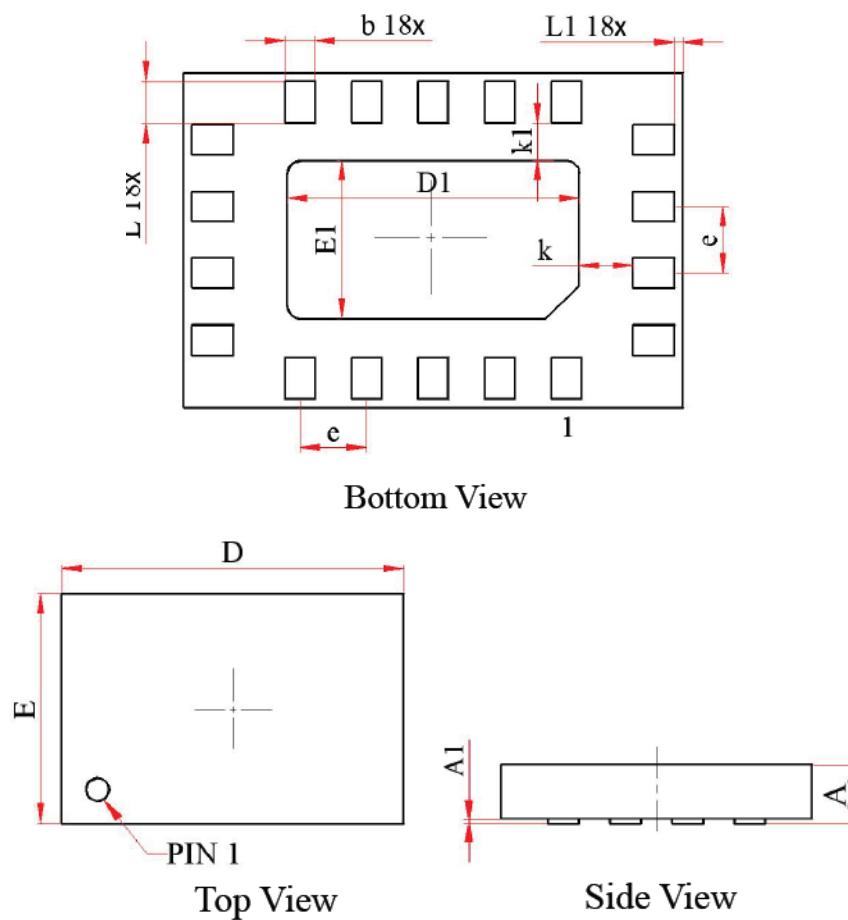


| Symbol | Dimension in Millimeters |      |      |
|--------|--------------------------|------|------|
|        | Min.                     | Typ. | Max. |
| A      | 0.50                     | 0.55 | 0.60 |
| A1     | 0.00                     | -    | 0.05 |
| A3     | 0.15 Typ.                |      |      |
| D      | 1.75                     | 1.80 | 1.85 |
| E      | 2.55                     | 2.60 | 2.65 |
| L      | 0.30                     | 0.40 | 0.50 |
| b      | 0.15                     | 0.20 | 0.25 |
| e      | 0.40 Typ.                |      |      |

# EMS4100

## Two-Channel Differential 2:1/1:2 USB 3.1 Super Speed 10Gbps Mux/DeMux

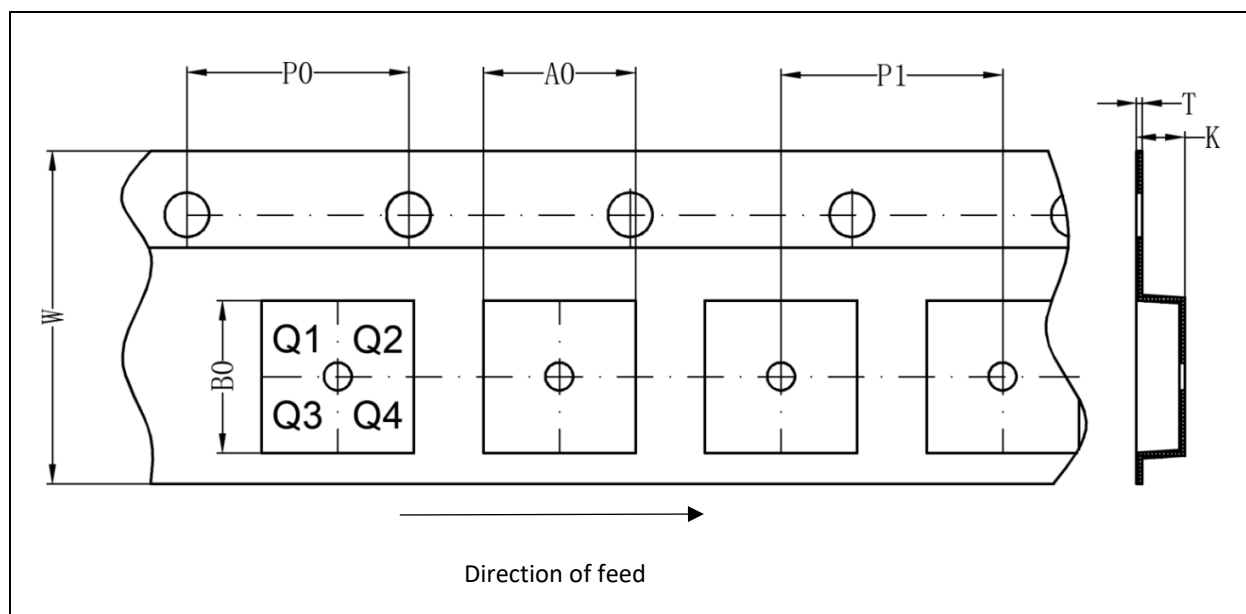
### QFN2x3-18L



| Symbol | Dimension In Millimeters |       |       | Dimension In Inches |       |       |
|--------|--------------------------|-------|-------|---------------------|-------|-------|
|        | Normal                   | Min   | Max   | Normal              | Min   | Max   |
| A      | --                       | 0.340 | 0.400 | --                  | 0.013 | 0.016 |
| A1     | --                       | 0.010 | 0.050 | --                  | 0.000 | 0.002 |
| D      | 3.000                    | 2.950 | 3.050 | 0.118               | 0.116 | 0.120 |
| E      | 2.000                    | 1.950 | 2.050 | 0.079               | 0.077 | 0.081 |
| D1     | 1.750                    | 1.700 | 1.800 | 0.069               | 0.067 | 0.071 |
| E1     | 0.950                    | 0.900 | 1.000 | 0.037               | 0.035 | 0.039 |
| b      | 0.180                    | 0.150 | 0.210 | 0.007               | 0.006 | 0.008 |
| L      | 0.250                    | 0.200 | 0.300 | 0.010               | 0.008 | 0.012 |
| L1     | 0.050                    | 0.010 | 0.090 | 0.002               | 0.000 | 0.004 |
| k      | 0.325 REF                |       |       | 0.013 REF           |       |       |
| k1     | 0.225 REF                |       |       | 0.009 REF           |       |       |
| e      | 0.400 BSC                |       |       | 0.016 BSC           |       |       |

## 13. Tape and Reel Information

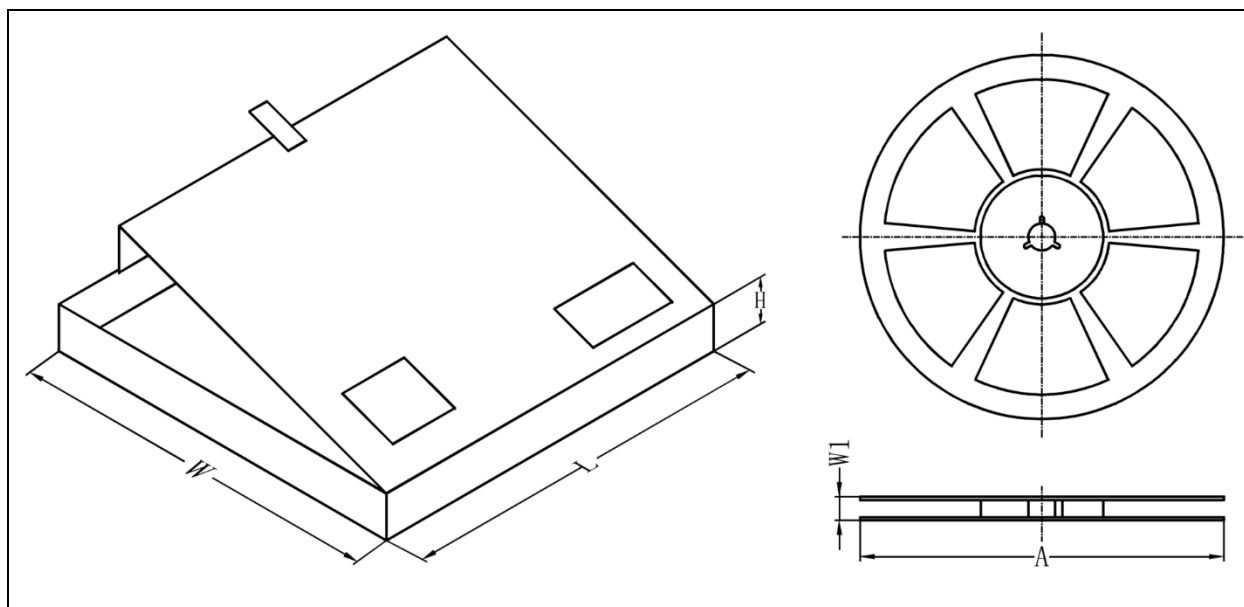
### 13.1. Carrier tape dimensions



**Table 6. Carrier tape dimensions**

| Package version | A0(mm) | B0(mm) | K(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|-------|-------|--------|-------|--------|-------|
| QFN2x3-18L      | 2.2    | 3.3    | 1.13  | 0.2   | 4      | 8     | 4      | Q1    |

## 13.2. Reel and box dimensions



**Table 7. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | SPQ (pcs)[1] | Reels per box | Outer box dimensions L×W×H(mm)[2] |
|-----------------|-----------------|-----------------------|--------------------|--------------|---------------|-----------------------------------|
| QFN2x3-18L      | 1R              | 180                   | 12.2               | 3000         | 1             | 210x200x40                        |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 14. Abbreviations

**Table 8. Abbreviations**

| Acronym | Description             |
|---------|-------------------------|
| ESD     | ElectroStatic Discharge |
| HBM     | Human Body Model        |

## 15. Revision History

**Table 9. Revision history**

| Document ID    | Release Date                                                                                                                                           | Data sheet status | Change notice | Supersedes     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|----------------|
| EMS4100 Rev1.4 | Dec 15, 2025                                                                                                                                           | Product datasheet |               | EMS4100 Rev1.3 |
| Modifications: | <ul style="list-style-type: none"> <li>Table 1: topside marking information updated.</li> <li>Fig.3: Top view pin configuration added EPAD.</li> </ul> |                   |               |                |
| EMS4100 Rev1.3 | Aug 30, 2025                                                                                                                                           | Product datasheet |               | EMS4100 Rev1.2 |
| Modifications: | <ul style="list-style-type: none"> <li>Table5: updated the spec of bandwidth.</li> </ul>                                                               |                   |               |                |
| EMS4100 Rev1.2 | Aug 27, 2025                                                                                                                                           | Product datasheet |               | EMS4100 Rev1.1 |
| Modifications: | <ul style="list-style-type: none"> <li>Added tape and reel information.</li> </ul>                                                                     |                   |               |                |
| EMS4100 Rev1.1 | Oct 30, 2024                                                                                                                                           | Product datasheet |               | EMS4100 Rev1.0 |
| Modifications: | <ul style="list-style-type: none"> <li>Added QFN2x3-18L package.</li> </ul>                                                                            |                   |               |                |
| EMS4100 Rev1.0 | Oct 30, 2023                                                                                                                                           | Product datasheet |               |                |